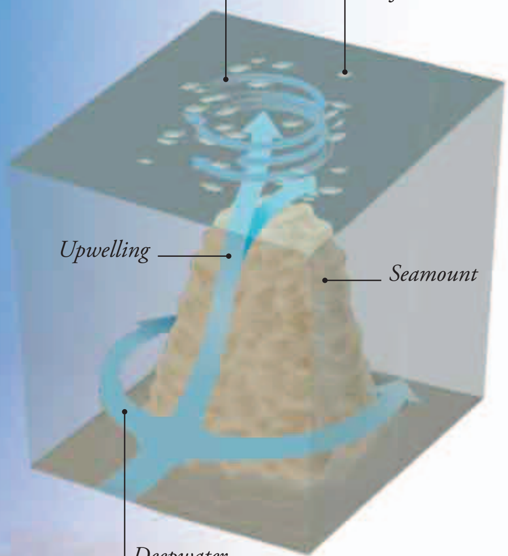
SEAMOUNTS

The submerged extinct volcanoes known as seamounts are covered with nutrient-rich sediments. Ocean currents flowing over the seamounts pick up the nutrients and carry them to the surface, creating local upwelling zones. These isolated hotspots often have their own unique wildlife.

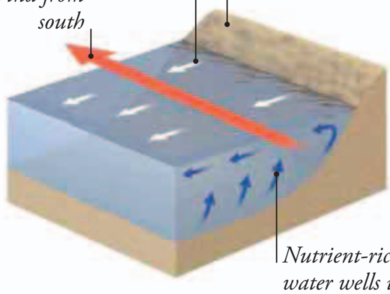


EL NIÑO

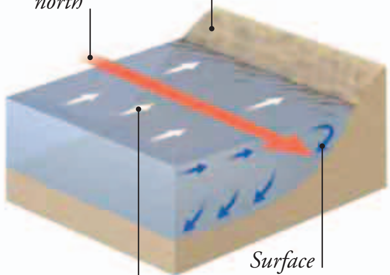
If upwelling stops, it has a big impact on ocean life. Sometimes the trade winds over the Pacific weaken, allowing warm surface water to flow east and smother an upwelling zone off tropical South America. Known as the El Niño effect, this stops the plankton growth, so the fish vanish—a disaster for fish-eating birds such as these blue-footed boobies.

HOW IT WORKS

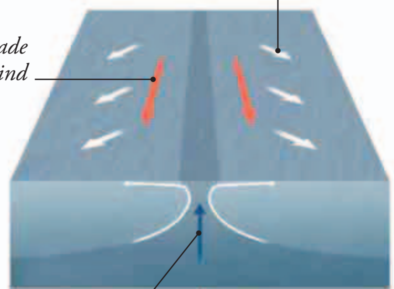
The Ekman transport effect can make strong winds blowing along the coast drag water away from the shore, creating an upwelling zone. Wind blowing in the opposite direction can cause downwelling. The pattern shown would be reversed in the northern hemisphere. The Ekman transport effect also draws surface water away from the equator, so cool, rich water wells up from below.



Coastal upwelling



Coastal downwelling



Equatorial upwelling